(1 Point)

$y' - y = e^{3x}$ *with* $y(0) = 0$. *Then* $y(1) =$

- ○ $\frac{e^3 - e}{2}$
- ○ $e^3 - e$
- ○ e^3
- ○ 2
- ○ *None of the other options*

3

(1 Point)

The minimum value of $f(x, y) = 4x - 2y$ subject to the condition $x^2 + y^2 = 5$ is attained at ____

- ○ $(-2, 1)$
- ○ $(2, 1)$
- ○ $(1, 2)$
- ○ $(\sqrt{2}, \sqrt{3})$
- ○ *None of the other options*

Question (1 Point)

$\sum_{n=1}^{\infty} (0.9)^n =$

Enter your math answer

5

Let the function f(x, y) be defined as follows: For (x, y) ≠ (0, 0):
f(x, y) = (x * y^3) / (x^3 + y^6). For (x, y) = (0, 0):
f(x, y) = 0. Then the directional derivative of f at the point (0, 0) in the direction of the vector (1, -1). (1 Point)

- -1
- 1
- 0
- None of the other options
- Does not exist

6

6

(1 Point)

The function $f(x, y) = 2 - x^2 - y^2$ has ——at $(0, 0)$.

- ◯ *Relative* max *imum*
- ◯ *Relative* min *imum*
- ◯ *Saddle nature*
- ◯ *Cannot be* det *er* min *ed*

7

Question (1 Point)

The sequence $f(n) = e^{-n} \sin 2n$ *for* $n \geq 1$

- ◯ *Converges to* 0
- ◯ *diverges*
- ◯ *Converges but will not tend to zero*
- ◯ *none of the other options*
- ◯ *Convergence cannot be established*

Question (1 Point)

The series $\sum_{n=1}^{\infty} (1 + e^{-n})$

- converges
- diverges
- converges absolutely
- converges conditionally
- None of the other options

9

Question (1 Point)

The series $\sum_{n=1}^{\infty} \frac{(-1)^n}{e^n}$

- Converges absolutely and is equal to $\frac{-1}{e+1}$
- Converges absolutely and is equal to $\frac{1}{e+1}$
- Converges conditionally and is equal to $\frac{-1}{e+1}$
- Converges conditionally and is equal to $\frac{1}{e+1}$
- Diverges